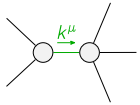


|                                                     |                           |                                               |                                                     |                                              |                                           |                                               |
|-----------------------------------------------------|---------------------------|-----------------------------------------------|-----------------------------------------------------|----------------------------------------------|-------------------------------------------|-----------------------------------------------|
|                                                     | $\mathcal{K}_{0^+}^{\#1}$ | $\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$ |                                                     |                                              |                                           |                                               |
| $\mathcal{K}_{0^+}^{\#1} \dagger$                   | 0                         | 0                                             |                                                     |                                              |                                           |                                               |
| $\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$ | 0                         | $3 k^2 \frac{(4)}{\kappa_{15}}$               | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$           | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$    | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$      | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$     | 0                         | 0                                             | 0                                                   | 0                                            | 0                                         | 0                                             |
| $\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$     | 0                         | 0                                             | 0                                                   | 0                                            | 0                                         | 0                                             |
| $\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha}$          | 0                         | 0                                             | $k^2 \frac{(4)}{\kappa_1}$                          | $-\frac{k^2 \frac{(4)}{\kappa_1}}{\sqrt{2}}$ |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha}$          | 0                         | 0                                             | $-\frac{k^2 \frac{(4)}{\kappa_1}}{\sqrt{2}}$        | $\frac{k^2 \frac{(4)}{\kappa_1}}{2}$         | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                     |                           |                                               | $\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$     | 0                                            | 0                                         |                                               |
|                                                     |                           |                                               | $\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$ | 0                                            | 0                                         |                                               |

|                                                     |                           |                                               |                                                     |                                                  |                                           |                                               |
|-----------------------------------------------------|---------------------------|-----------------------------------------------|-----------------------------------------------------|--------------------------------------------------|-------------------------------------------|-----------------------------------------------|
|                                                     | $\mathcal{J}_{0^+}^{\#1}$ | $\mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi}$ |                                                     |                                                  |                                           |                                               |
| $\mathcal{J}_{0^+}^{\#1} \dagger$                   | 0                         | 0                                             |                                                     |                                                  |                                           |                                               |
| $\mathcal{J}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$ | 0                         | $\frac{1}{3 k^2 \frac{(4)}{\kappa_{15}}}$     | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$           | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$        | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha}$      | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$     | 0                         | 0                                             | 0                                                   | 0                                                | 0                                         | 0                                             |
| $\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$     | 0                         | 0                                             | 0                                                   | 0                                                | 0                                         | 0                                             |
| $\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha}$          | 0                         | 0                                             | $\frac{4}{9 k^2 \frac{(4)}{\kappa_1}}$              | $-\frac{2 \sqrt{2}}{9 k^2 \frac{(4)}{\kappa_1}}$ |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha}$          | 0                         | 0                                             | $-\frac{2 \sqrt{2}}{9 k^2 \frac{(4)}{\kappa_1}}$    | $\frac{2}{9 k^2 \frac{(4)}{\kappa_1}}$           | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                     |                           |                                               | $\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$     | 0                                                | 0                                         |                                               |
|                                                     |                           |                                               | $\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$ | 0                                                | 0                                         |                                               |

### Lagrangian

$$\begin{aligned}
& \kappa_1^{(4)} \partial_\beta \mathcal{K}_{\chi\delta}^\delta \partial^\chi \mathcal{K}_\alpha^{\beta-} - \kappa_1^{(4)} \partial_\chi \mathcal{K}_{\beta\delta}^\delta \partial^\chi \mathcal{K}_\alpha^{\beta+} + 2 \kappa_{15}^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta - 4 \kappa_{15}^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \\
& 2 \kappa_{15}^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - \kappa_{15}^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \kappa_{15}^{(4)} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \kappa_{15}^{(4)} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

|                                                                                    |                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                             |
|------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Added source term(s):                                                              | $\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                             |
| Source constraint(s)                                                               | # constraint(s)                                               | Covariant form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             |
| $\mathcal{J}_{0^+}^{\#1} == 0$                                                     | 1                                                             | $\partial_\beta \mathcal{J}^{\alpha\beta} = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             |
| $\mathcal{J}_{1^+}^{\#1\alpha} + 2 \mathcal{J}_{1^+}^{\#2\alpha} == 0$             | 3                                                             | $\partial_\chi \partial^\alpha \mathcal{J}_{\beta}^{\beta\chi} + \partial_\chi \partial^\chi \mathcal{J}_{\beta}^{\beta\alpha} == 3 \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                             |
| $\mathcal{J}_{1^+}^{\#2\alpha\beta} == 0$                                          | 3                                                             | $\partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} == \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                             |
| $\mathcal{J}_{1^+}^{\#1\alpha\beta} == 0$                                          | 3                                                             | $\partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} == \partial_\delta \partial_\chi \partial^\delta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} == 0$                                      | 5                                                             | $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$<br>$4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}{}_\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha}{}_\delta ==$<br>$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$<br>$2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}{}_\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta}{}_\delta$ |                             |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$                                          | 5                                                             | $3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^{\chi}{}_\chi{}^\delta == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^{\chi}{}_\chi{}^\delta + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                             |
| Total # constraint(s):                                                             | 20                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                             |
| Resolved pole(s)                                                                   | # polarization(s)                                             | Square mass                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Residue                     |
|  | 2                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $-\frac{1}{\kappa_1^{(4)}}$ |
| Resolved unitarity condition(s):                                                   | $\kappa_1^{(4)} < 0$                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                             |